Lab Exercise 19 - ClusterIP in Service Kubernetes

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Batch:- DevOps - B1

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A ClusterIP is the default type of service in Kubernetes. It provides a stable internal IP address for the service, which is accessible only within the cluster. This is useful for internal communication between services within the cluster.

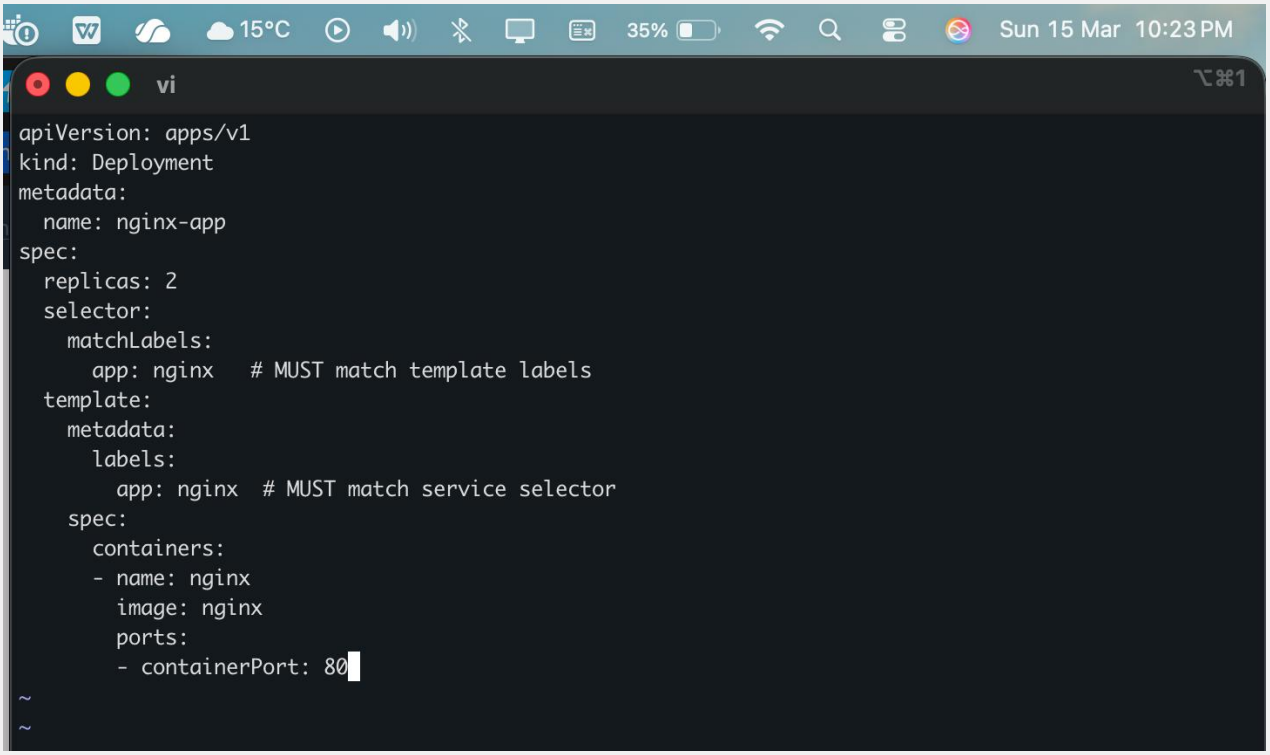
Step 1: Create Deployment (Correct Labels)

Create file:

nginx-deployment.yaml

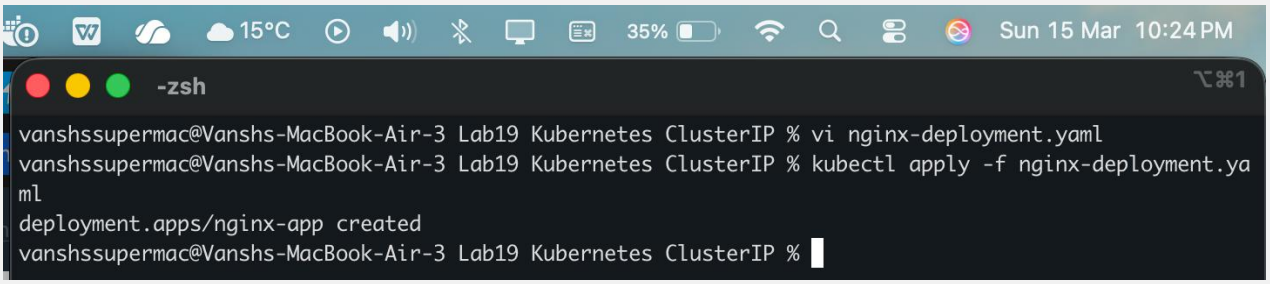
```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-app
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx   # MUST match template labels
  template:
    metadata:
      labels:
```

```
    app: nginx  # MUST match service selector
  spec:
   containers:
   - name: nginx
     image: nginx
     ports:
     - containerPort: 80
```



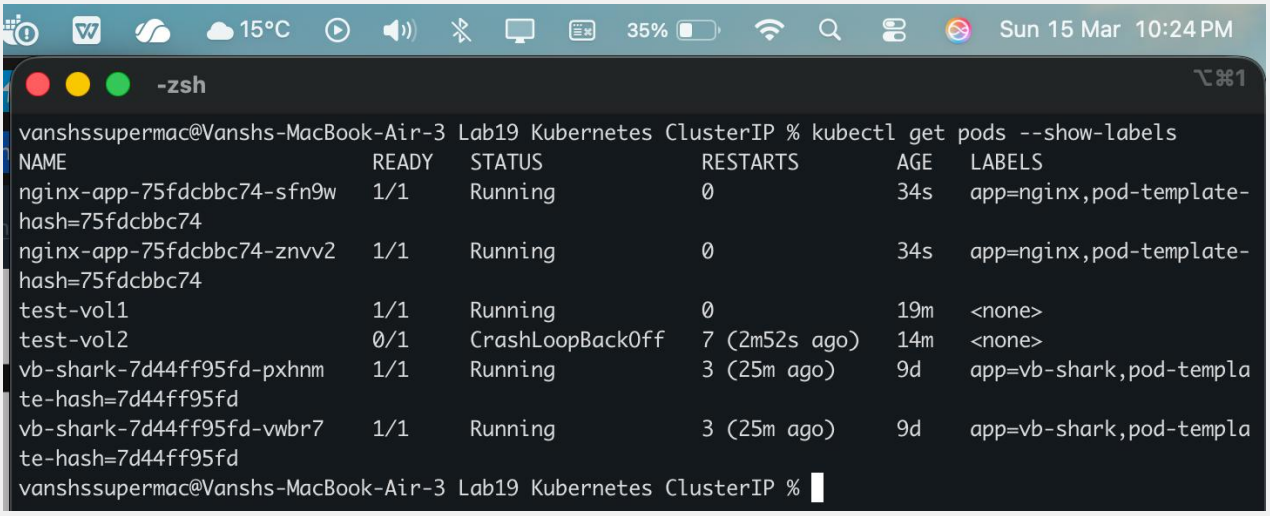
Save file.

```
kubectl apply -f nginx-deployment.yaml
```



Verify:

```
kubectl get pods --show-labels
```



You should see:

```
app=nginx
```

Step 2: Create ClusterIP Service (Matching Selector)

Create file:

nginx-service.yaml

```
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  type: ClusterIP
  selector:
    app: nginx   # MUST match pod label
  ports:
```

```
- port: 80
  targetPort: 80
```

```
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  type: ClusterIP
  selector:
    app: nginx   # MUST match pod label
  ports:
  - port: 80
    targetPort: 80
~
~
```

Apply:

```
kubectl apply -f nginx-service.yaml
```

```
vanshssupermac@Vanshs-MacBook-Air-3 Lab19 Kubernetes ClusterIP % vi nginx-service.yaml
vanshssupermac@Vanshs-MacBook-Air-3 Lab19 Kubernetes ClusterIP % kubectl apply -f nginx-service.yaml
service/nginx-service created
vanshssupermac@Vanshs-MacBook-Air-3 Lab19 Kubernetes ClusterIP %
```

Step 3: Verify It Is Working

Check Service

```
kubectl get svc
```

```
vanshssupermac@Vanshs-MacBook-Air-3 Lab19 Kubernetes ClusterIP % kubectl get svc
NAME            TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)        AGE
kubernetes      ClusterIP   10.96.0.1        <none>        443/TCP        34d
nginx-service   ClusterIP   10.103.215.137   <none>        80/TCP         26s
vb-python       NodePort    10.101.7.9       <none>        80:30007/TCP   7d2h
vanshssupermac@Vanshs-MacBook-Air-3 Lab19 Kubernetes ClusterIP %
```

You should see:

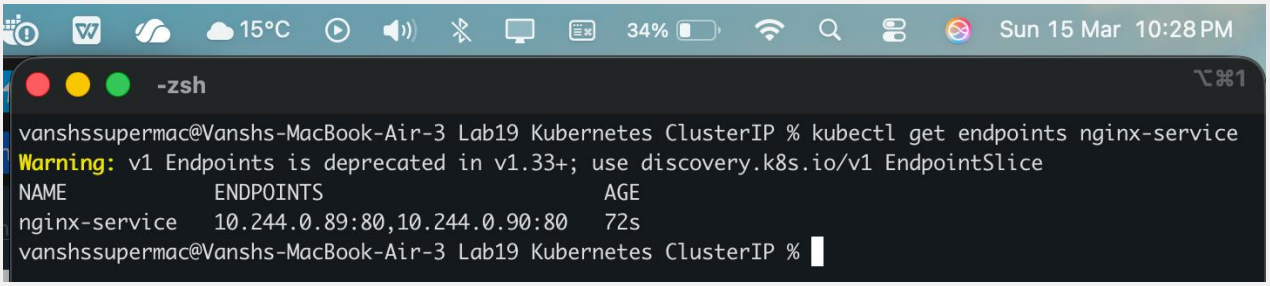
```
TYPE       CLUSTER-IP
ClusterIP  10.x.x.x
```

MOST IMPORTANT CHECK

```
kubectl get endpoints nginx-service
```

Now you should see something like:

```
10.244.x.x:80, 10.244.x.x:80
```



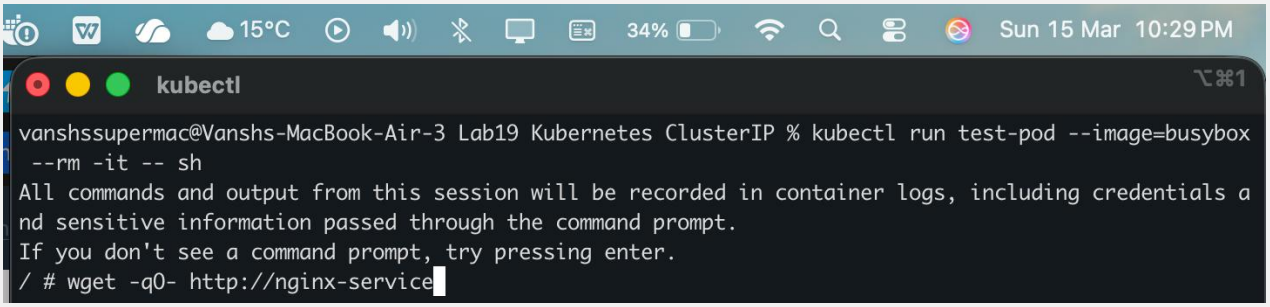
Step 4: Test From Inside Cluster

Run test pod:

```
kubectl run test-pod --image=busybox --rm -it -- sh
```

Inside pod:

```
wget -qO- http://nginx-
service
```



You should see:

Welcome to

nginx!

```
vanshssupermac@Vanshs-MacBook-Air-3 Lab19 Kubernetes ClusterIP % kubectl run test-pod --image=busybox
 --rm -it -- sh
All commands and output from this session will be recorded in container logs, including credentials a
nd sensitive information passed through the command prompt.
If you don't see a command prompt, try pressing enter.
/ # wget -qO- http://nginx-service
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br/>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br/>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
/ #
```

Exit:

Exit

```
</html>
/ # exit
Session ended, resume using 'kubectl attach test-pod -c test-pod -i -t' command when the pod is runni
ng
pod "test-pod" deleted from default namespace
vanshssupermac@Vanshs-MacBook-Air-3 Lab19 Kubernetes ClusterIP %
```

Step 5: Test From Outside (It Will Fail)

From your local machine:

curl http://<CLUSTER-IP>

```
15°C  34%  Sun 15 Mar 10:32 PM
curl
vanshssupermac@Vanshs-MacBook-Air-3 Lab19 Kubernetes ClusterIP % curl http://10.103.215.137
```

It will NOT work.

That proves:

ClusterIP works only inside cluster.

Thank You